

MH1100 Calculus I

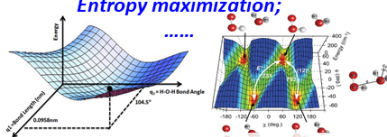
Handout 9 - Application of Differentiation I

Application of Differentiation

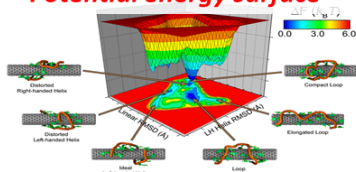
- Maximum and Minimum Values
- The Mean Value Theorem

Further readings:

*Optimization;
Extreme value theory;
Energy minimization;
Entropy maximization;*



Potential energy surface



❖ **Maximum and Minimum Values**

Further readings:

Mathematical analysis:

*differentiation, integration, measure,
infinite series, analytic functions, etc*

Real analysis:

*convergence, limits, continuity,
smoothness, differentiability,
integrability, etc;*

Complex analysis;

Functional analysis;

Differential equations;

Measure theory;

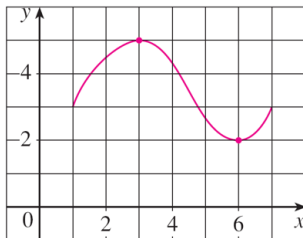
Numerical analysis;

.....

❖ **The Mean Value Theorem**

Maximum and Minimum Values

Let's first explain exactly what we mean by maximum and minimum values. We see that the highest point on the graph of the function $f(x)$ shown in the figure is the point $(3, 5)$.



domain: $[1, 7]$

In other words, the largest value of $f(x)$ is $f(3) = 5$. Likewise, the smallest value is $f(6) = 2$.

We say that $f(3) = 5$ is the absolute maximum of $f(x)$ and $f(6) = 2$ is the absolute minimum. *Not very precise ... ;)*

Maximum and Minimum Values

In general, we use the following definition

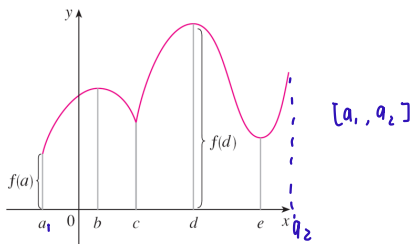
Let c be a number in the domain D of a function $f(x)$. Then $f(c)$ is the

- **absolute maximum** value of $f(x)$ on D , if $f(c) \geq f(x)$ for all x in D .
- **absolute minimum** value of $f(x)$ on D , if $f(c) \leq f(x)$ for all x in D .

An absolute maximum or minimum is sometimes called a **global** maximum or minimum. The maximum and minimum values of $f(x)$ are called **extreme values** of $f(x)$.

Maximum and Minimum Values

The figure below shows the graph of a function $f(x)$ with absolute maximum at $x = d$ and absolute minimum at $x = a$. Note that $(d, f(d))$ is the highest point on the graph and $(a, f(a))$ is the lowest point.



If we consider values of x near b (for instance, if we restrict our attention to the interval (a, c)), then $f(b)$ is the largest values in the interval and is called a **local maximum value** of $f(x)$.

Likewise, $f(c)$ is called a **local minimum value** of $f(x)$. The function $f(x)$ also has a local minimum at $x = e$.

Maximum and Minimum Values

In general, we have the following definition.

The number $f(c)$ is a

- **local maximum** value of $f(x)$ if $f(c) \geq f(x)$ when x is near c .
- **local minimum** value of $f(x)$ if $f(c) \leq f(x)$ when x is near c .

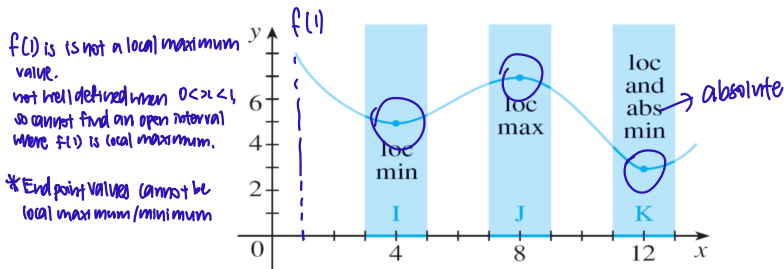
To be more specific, if we say that something is true near c , we mean that it is true on **some open interval containing c** .

there exists $h, \alpha \in (c-h, c+h)$, $f(c) \geq f(x) \rightarrow f(c)$ is local maximum

$f(c) \leq f(x) \rightarrow f(c)$ is local minimum

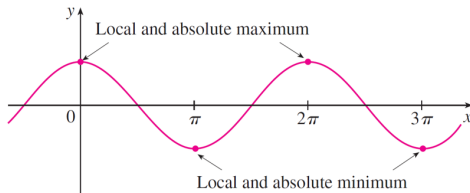
Maximum and Minimum Values

For instance, in the figure below, we see that $f(4) = 5$ is a local minimum because it is the smallest value of f on the interval I . It is not the absolute minimum, because $f(x)$ takes smaller values when x is near 12. In fact $f(12) = 3$ is both a local minimum and the absolute minimum. Similarly, $f(8) = 7$ is a local maximum, but not the absolute maximum because $f(x)$ takes larger values near 1.



Maximum and Minimum Values

For function $f(x) = \cos x$, it has **infinitely many** both local and absolute maximum values $f = 1$, since $\cos(2n\pi) = 1$ for any integer n and $-1 \leq \cos x \leq 1$ for all x .



Since $\cos(2n + 1)\pi = -1$, function $f(x) = \cos x$ also has infinitely many both local and absolute minimum values $f = -1$.

Maximum and Minimum Values

The following theorem gives conditions under which a function is guaranteed to possess extreme values.

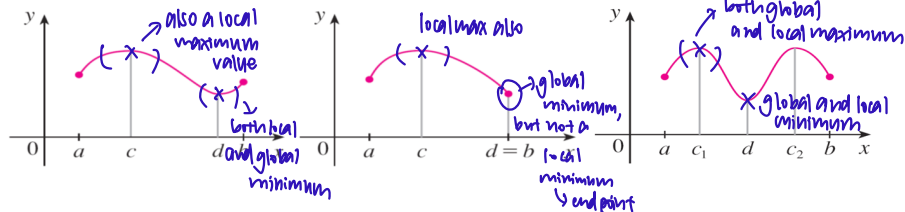
2 conditions

The Extreme Value Theorem

- ① If $f(x)$ is continuous on a closed interval $[a, b]$, then $f(x)$ attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some number c and d in $[a, b]$.

The Extreme Value Theorem is illustrated in the figure below,

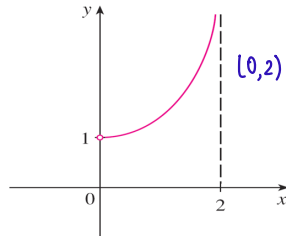
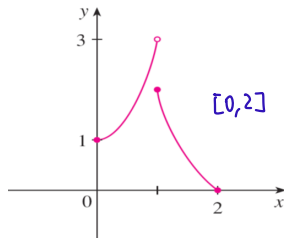
global maximum



Again, a function can attain extreme values more than once.

Maximum and Minimum Values

Figures below show that a function may not have extreme values, if either hypothesis (continuity or closed interval) is omitted from the Extreme Value Theorem.

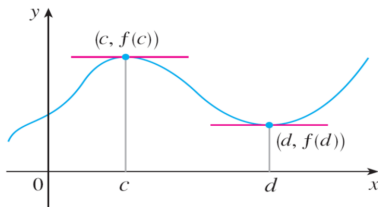


Left Figure: This function has no maximum value. The function is **not continuous** on the interval. The range of $f(x)$ is $[0, 3)$, it can never attain the value 3.

Right Figure: This function has neither a maximum nor a minimum value. The function is continuous on the **open interval** $(0, 2)$, and the range of the function is $(1, \infty)$.

Maximum and Minimum Values

The Extreme Value Theorem does not tell us how to find these extreme values. However, we can start by looking for local extreme values.



The above figure shows the function $f(x)$ has a local maximum at $x = c$ and a local minimum at $x = d$. It appears that at the maximum and minimum points, the tangent lines are horizontal thus have slope 0. We know that the slope of the tangent line is exactly the derivative at that point, thus we should have $f'(c) = 0$ and $f'(d) = 0$.

Maximum and Minimum Values

Fermat's Theorem

If f has a local maximum or minimum at c , and if $f'(c)$ exists, then $f'(c) = 0$.

Prove: Suppose f is a local minimum at c . From its definition, there exists h , when x is in the domain $(c-h, c+h)$, we have $f(x) \geq f(c)$. Therefore,

$$\frac{f(x) - f(c)}{x - c} \geq 0 \quad \text{for } (x > c);$$

$c < x < c+h$

$$\frac{f(x) - f(c)}{x - c} \leq 0 \quad \text{for } (x < c);$$

$c-h < x < c$

$$\lim_{x \rightarrow c^+} \frac{f(x) - f(c)}{x - c} \geq 0;$$

$$\lim_{x \rightarrow c^-} \frac{f(x) - f(c)}{x - c} \leq 0$$

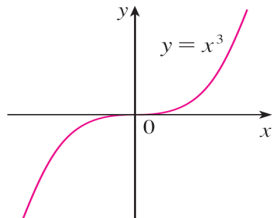
Since the limit exists at $x = c$, so we have $f'(c) = 0$.

The same idea can be used to prove the result for local maximum.

Maximum and Minimum Values

→ cannot guarantee it is always a local extreme value
If $f'(c) = 0$, can we have $f(x)$ takes either local minimum or maximum at $x = c$?

For example, $f(x) = x^3$ at $x = 0$. We have $f'(x) = 3x^2$ and $f'(0) = 0$, but $f(x)$ has no maximum or minimum at 0 as show in the figure below.



$f(0)$ in this case can never be the local extreme value.

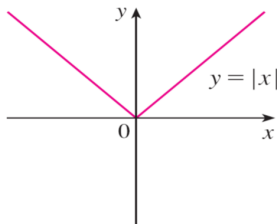
The condition $f'(0) = 0$ simply means that the curve $y = x^3$ has a horizontal tangent at $(0,0)$.

Maximum and Minimum Values

If $f(x)$ takes a local minimum or maximum at $x = c$, can we have $f'(c) = 0$?

For example, function $f(x) = |x|$ has its (local and absolute) minimum value at $x = 0$, but $f'(0)$ does not even exist.

do not even check its values



There may be an extreme value even when $f'(c)$ does not exist. The above examples show that we must be careful when using Fermat's Theorem. In general, the converse of Fermat's Theorem is false.

Maximum and Minimum Values

However, Fermat's Theorem and the above examples do suggest that we should start looking for extreme values of $f(x)$ at number c , where $f'(c) = 0$ or where $f'(c)$ does not exist. Such numbers have a special name.

A **critical number** of a function $f(x)$ is a number c in the domain of $f(x)$ such that either $\underline{f'(c) = 0}$ or $\underline{f'(c)}$ does not exist.

②

①

In terms of critical numbers, Fermat's Theorem can be rephrased as follows.

If $f(x)$ has a local maximum or minimum at $x = c$, then c is a critical number of $f(x)$.

* if have critical point, does not imply it is an extreme value.

① $f'(c)$ D.N.E $\Rightarrow c$ is a critical no. of $f(x)$

② $f'(c)$ exists $\Rightarrow f'(c) = 0 \Rightarrow c$ is a critical no. of $f(x)$.

Maximum and Minimum Values

Now we are ready to find the absolute maximums or minimums of a continuous function on a closed interval.

We note that absolute maximums (or minimums) occur only at local maximums (or minimums) or at endpoints of the interval, thus the following three-step procedure always works.

The Closed Interval Method

To find the absolute maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

Step 1. Find the values of f at the critical numbers of f in (a, b) .

Step 2. Find the values of f at the endpoints of the interval.

Step 3. The largest of these values from Step 1 and 2 is the absolute maximum values; the smallest of these values is the absolute minimum value.

Understanding Checker

1. Find the second derivative y'' if $x^4 + y^4 = 16$

(A) $-48 \frac{x^4}{y^3}$ (B) $48 \frac{x^4}{y^3}$ (C) $-\frac{x^4}{y^3}$ (D) $-3x^2 \frac{y - xy'}{y^3}$

$$4x^3 + 4y^3 y' = 0$$

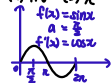
$$y' = -\frac{x^3}{y^3}$$

$$y'' = -\frac{y^2(3x^2) - x^3(3y^2)y'}{y^6}$$

$$= -3x^2 \frac{y - xy'}{y^4} = -3x^2 \cdot \frac{y + x \cdot \frac{x^3}{y^3}}{y^4} = -3x^2 \cdot \frac{y^4 + x^4}{y^7} = -\frac{48x^2}{y^7} \therefore (A)$$

2. Use linear approximation to calculate $\sin(91^\circ)$.

(A) 1.0 (B) $\frac{1}{e}$ (C) $-\frac{1}{e}$ (D) -1.0



$$91^\circ = \frac{91}{180} \pi \quad \frac{1}{\pi} \approx 0.3$$

$$\approx \frac{\pi}{2}$$

Linear approximation:

$$f(x) \approx L(x) = f(a) + f'(a)(x-a)$$

$$\sin(91^\circ) \approx \sin \frac{\pi}{2} + \cos \left(\frac{\pi}{2} \right) \left(\frac{91}{180} \pi - \frac{\pi}{2} \right)$$

$$= 1 + 0 \cdot \left(\frac{91}{180} \pi - \frac{\pi}{2} \right) = 1$$

3. Find the limit $\lim_{x \rightarrow 1} \frac{x^{17} - 1}{x - 1}$

(A) 0 (B) D.N.E (C) 17 (D) 16

$$(x^2 - 1) = (x-1)(1+x)$$

$$(x^3 - 1) = (x-1)(1+x+x^2)$$

$$(x^n - 1) = (x-1)(1+x+x^2+\dots+x^{n-1})$$

$$\lim_{x \rightarrow 1} \frac{(x-1)(1+x+\dots+x^{16})}{x-1} = \lim_{x \rightarrow 1} (1+x+\dots+x^{16}) = 17$$

$$f(x) = x^{17}$$

$$\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = f'(x) \Big|_{x=1}$$

$$= 17x^{16} \Big|_{x=1}$$

$$= 17$$

4. Find global maximum and local maximum for function $f(x) = 3$ defined on domain $[1, 5]$.

(A) 3 and 3 (B) 3 and D.N.E (C) D.N.E and 3 (D) D.N.E and D.N.E

$$f(x) \leq 3 \quad x \in [1, 5]$$

(A)

Maximum and Minimum Values

Example

Find the absolute maximum and minimum values of the function

$$f(x) = x^3 - 3x^2 + 1 \quad -1/2 \leq x \leq 4.$$

Solution Find critical numbers,

$$f'(x) = 3x^2 - 6x = \underbrace{3x(x-2)}_0$$

check that the
critical numbers are
in the domain

Since $f'(x)$ exists for all x , the only critical numbers are $x = 0$ and $x = 2$. We have $f(0) = 1$ and $f(2) = -3$.

The values of $f(x)$ at the endpoints are $f(-1/2) = 1/8$ and $f(4) = 17$.

Comparing these four values, we can see that the absolute maximum value is $f(4) = 17$, and the absolute minimum value is $f(2) = -3$.

3.2 The Mean Value Theorem

Mean Value Theorem is essential to Calculus. But to arrive at the Mean Value Theorem, we need the following result.

Rolle's Theorem: Let f be a function that satisfies the following three hypotheses:

- 1 f is continuous on the closed interval $[a, b]$. $\Rightarrow f(x)$ has global max/min
- 2 f is differentiable on the open interval (a, b) .
- 3 $f(a) = f(b)$. $\Rightarrow$ if $f(a)$ is global max, $f(b)$ cannot be global min (unless they have constant function) $\Rightarrow c \in (a, b)$, $f(c)$ global min

Then there is a number c in (a, b) , such that $f'(c) = 0$.

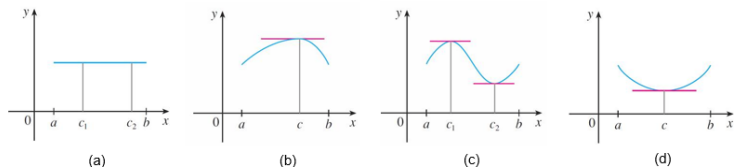
$f(c)$ local min

$\Downarrow$

$f'(c) = 0$

if $f(a)$ is global min,
 $c \in (a, b)$, $f(c)$ global max
 $\Downarrow$
 $f(c)$ local max

The Mean Value Theorem



Case I $f(x) = k$, a constant. [Figure (a)]

Then $f'(x) = 0$, so the number c can be taken as any number in (a, b) .

Case II $f(x) > f(a)$, for some x in (a, b) . [Figure (b) and (c)]

By Extreme Value Theorem, f has a global maximum value in $[a, b]$. Since $f(a) = f(b)$, it must attain this global maximum value at a number c in the open interval (a, b) . Then f has a local maximum at c , f is also differentiable at c . Therefore $f'(c) = 0$ by Fermat's Theorem.

Case III $f(x) < f(a)$, for some x in (a, b) . [Figure (c) and (d)]

Same as Case II.

The Mean Value Theorem

Example

Prove that the equation $x^3 + x - 1 = 0$ has exactly one real root.

Solution First we use the Intermediate Value Theorem to show that a root exists. Let $f(x) = x^3 + x - 1$. Then $f(0) = -1 < 0$ and $f(1) = 1 > 0$.

Since f is a polynomial, it is continuous, so the Intermediate Value

Theorem states that there is a number c between 0 and 1 such that $f(c) = 0$. Thus the given equation has a root.

*polynomial function,
always continuous.*

The Mean Value Theorem

To show that the equation has no other real root or roots, we use Rolle's Theorem and argue by contradiction.

Suppose that it had two roots a and b , then $f(a) = f(b) = 0$. Since f is a polynomial, it is differentiable on (a, b) and continuous on $[a, b]$. Thus, by Rolle's Theorem, there is a number c between a and b such that $f'(c) = 0$.
 $c \in (a, b)$

But $f'(x) = 3x^2 + 1 \geq 1$ for all x (since $x^2 \geq 0$), so $f'(x)$ can never be 0. This gives a contradiction.

Therefore the equation can't have more than one real root. That is the equation has exactly one real root.

The Mean Value Theorem

Rolle's Theorem can be directly used to prove the Mean Value Theorem.

MVT

The Mean Value Theorem: Let f be a function that satisfies the following hypotheses:

- 1 f is continuous on the closed interval $[a, b]$.
- 2 f is differentiable on the open interval (a, b) .

Rolle's Theorem has 1 extra condition

$$\textcircled{3} \begin{aligned} f(b) &= f(a) \\ f'(c) &= 0 \end{aligned}$$

Then there is a number c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

or, equivalently

$$\underbrace{f(b) - f(a)}_{\text{function}} = \underbrace{f'(c)}_{\text{derivative}} (b - a)$$

The Mean Value Theorem

Before proving this theorem, we can interpret its geometrical meaning.

The slope of the secant line AB is $m_{AB} = \frac{f(b)-f(a)}{b-a}$.

The Mean Value Theorem indicates there exists a point $(c, f(c))$, and $f'(c) = \frac{f(b)-f(a)}{b-a}$.

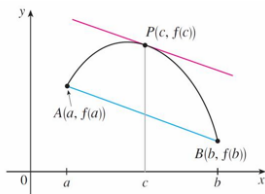


Figure 3

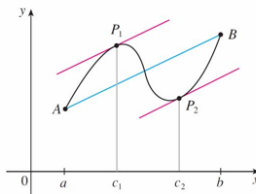


Figure 4

Imagine fix the curve (by iron metal), and rotate point B counterclockwise until $f(a) = f(b)$, i.e., $m_{AB} = 0$. From Rolle's theorem, we will have $f'(c) = 0$, i.e., $f'(c) = m_{AB}$.

Prove of the Mean Value Theorem

We define a new function,

$$h(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a)$$

Handwritten notes: "constant" with an arrow pointing to $f(a)$; "1st order polynomial" with an arrow pointing to the fraction term.

The new function satisfies $h(a) = h(b) = 0$, and $h'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$

We can use the Rolle's Theorem, as we have

- (1) $h(x)$ is continuous on $[a, b]$;
 - (2) $h(x)$ is differentiable on (a, b) ;
 - (3) $h(a) = h(b)$
- Handwritten note:* $\left. \begin{array}{l} (1) \\ (2) \end{array} \right\} \exists c \in (a, b), f'(c) = 0$

Therefore we can find a c in (a, b) , such that $h'(c) = 0$, which is to say

$$0 = h'(c) = f'(c) - \frac{f(b) - f(a)}{b - a},$$

so we have

$$f'(c) = \frac{f(b) - f(a)}{b - a} \quad \text{or} \quad f(b) - f(a) = f'(c)(b - a)$$

The Mean Value Theorem

To illustrate the Mean Value Theorem, we can consider

$$f(x) = x^3 - x, \quad a = 0, \quad b = 2.$$

Since $f(x)$ is a polynomial, it is continuous and differentiable for all x , so it is certainly continuous on $[0, 2]$ and differentiable on $(0, 2)$. Therefore, by the Mean Value Theorem, there is a number c in $(0, 2)$ such that

$$f(2) - f(0) = f'(c)(2 - 0)$$

Now $f(2) = 6$, $f(0) = 0$ and $f'(x) = 3x^2 - 1$, we have

$$6 - 0 = (3c^2 - 1) 2 = 6c^2 - 2$$

which gives $c^2 = \frac{4}{3}$, that is, $c = \pm \frac{2}{\sqrt{3}}$. But c must lie in the region $(0, 2)$, so $c = \frac{2}{\sqrt{3}}$.

The Mean Value Theorem

Figure 6 illustrates this calculation:

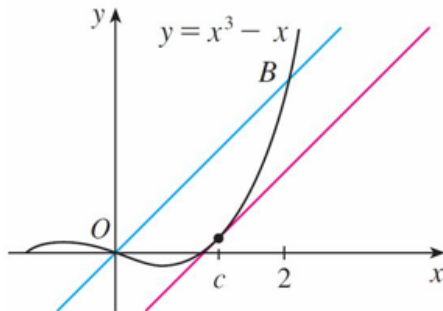


Figure 6

The tangent line at this value of c is parallel to the secant line OB .

The Mean Value Theorem

Example

Suppose that $f(0) = -3$ and $f'(x) \leq 5$ for all values of x . *→ differentiable everywhere* How large can $f(2)$ possibly be?

Solution As f is differentiable (and therefore continuous) everywhere, Mean Value Theorem can be used on the interval $[0, 2]$. There exists a number c such that:

$$f(2) - f(0) = f'(c)(2 - 0)$$

$$f(2) = f(0) + 2f'(c) = -3 + 2f'(c)$$

We have $f'(x) \leq 5$ for all x , so we have $f'(c) \leq 5$, thus $2f'(c) \leq 10$. And we have

$$f(2) = -3 + 2f'(c) \leq -3 + 10 = 7$$

The largest possible value for $f(2)$ is 7.

The Mean Value Theorem

Theorem If $f'(x) = 0$ for all x in an interval (a, b) , then f is constant on (a, b) .

Prove Let x_1 and x_2 be any two numbers in (a, b) and $x_1 < x_2$. Since f is differentiable on (a, b) , it must be differentiable on (x_1, x_2) and continuous on $[x_1, x_2]$. By the Mean Value Theorem, we have a number c , such that c is in (x_1, x_2) and

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

Since $f'(x) = 0$ for all x , we have $f'(c) = 0$, and

$$f(x_2) - f(x_1) = 0 \quad \text{or} \quad f(x_2) = f(x_1)$$

Therefore f has the same value at any two numbers x_1 and x_2 . This means that f is constant on (a, b) .

The Mean Value Theorem

Corollary If $f'(x) = g'(x)$ for all x in an interval (a, b) , then $f - g$ is constant on (a, b) ; that is, $f(x) = g(x) + c$, where c is constant.

Prove Let $F(x) = f(x) - g(x)$, then

$$F'(x) = f'(x) - g'(x) = 0$$

for all x in (a, b) .

Thus, from the above Theorem, $F(x)$ is a constant. Therefore $f(x) - g(x)$ is constant.

The Mean Value Theorem

Note

Care must be taken in applying the above Theorem. For example,

$$f(x) = \frac{x}{|x|} = \begin{cases} 1 & \forall x > 0 \\ -1 & \forall x < 0 \end{cases}$$

The domain of f is $D = \{x \mid x \neq 0\}$ and $f'(x) = 0$ for all x in D . But f is obviously not a constant function.

This does not contradict the Theorem, because D is not an interval!!
Notice that f is constant on the interval $(0, \infty)$ and also on the interval $(-\infty, 0)$.